



Staying in touch with presence

NOKIA
CONNECTING PEOPLE

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Introduction

The mobile phone is exceedingly evolving into a personal device, even to the extent of becoming part of the user's identity – this applies to the look and feel of the terminal itself as well as to how and why it is used. Multimedia messaging, for example, enables person-to-person communication to be richer and carry more emotional content, while downloadable icons and ringing tones along with changeable phone covers give users the ability to express themselves in a fashionable way. Applications and services that do the same and support users' natural communication patterns and individual needs in a fun and efficient way will be the most popular and desirable.

Presence will be one of these applications.

How Presence differs from instant messaging

Presence is a familiar concept to those using instant messaging services on the fixed Internet. With mobile phones, presence will not only enhance messaging but will introduce a service of its own that can be used in many other applications and services. It will be at the centre of all communication and mobile telephony will benefit from presence information just as it has done from messaging.

It will also be an appealing business opportunity to both mobile operators and service providers, opening more enriched applications with a personal look and feel that will result in increased revenue and reduced customer churn. A presence service is a tool allowing users to create their own content and share it with others and makes not just technology, but also services and communication itself more sensitive and personal.

According to another definition of presence, it is a dynamic variable profile of the user, which is visible to others and used to represent oneself, share information and control services. In essence, presence is two things: a user's status to others and others' status to the user. Status may contain information such as personal and device status, location or context, terminal capabilities and preferred contact method.

Benefiting from Presence

Presence supports the need of users to continuously share their availability, whereabouts, intentions, preferences and even emotions. Even today, people would like to know what their friends and colleagues are doing before trying to contact them. In this sense, presence

supports users' rational needs to communicate. Presence information is also personal: it is always linked to a particular person.

The concept of presence enables a delicate balance of communication. Firstly, it shows the person initiating the communication whether the other person is available and willing to communicate. On the other hand, presence information can be used to communicate to others when a person is able and willing to communicate as well as with whom and by what means. This will allow users to control their own communication more effectively.

Availability sharing may raise some security and privacy concerns. These are delicate issues and there are cultural differences between how people react towards them. In short, users should control their own specific presence information and have the final say on how it is used.

There will also be many different user groups, which will use presence information for different purposes. These groups will range from corporate business users to today's youth segment. While in corporate usage presence is likely to be used more for rational availability management. Young consumers are constantly seeking new ways to express themselves and build identity in a fun and visually rich way. Successful presence services will be adaptable to the different needs of varied user groups and segments.

Although presence is tightly bound to a human context, things and places may also have presence. And furthermore, network elements may use the presence information of people in creating and providing more personalised services as well as

deducing people's presence from raw presence data provided by different sources.

Putting presence in the mobile context

In the mobile domain, presence will be a richer and even more powerful service than we have become used to in the fixed Internet. The mobile context will expand presence from being online for instant messaging into a powerful tool of self-expression and continuous mobile awareness.

The reasons why presence will enhance mobile communications and provide an appealing service are easy to find: the mobile terminal is already often the primary storage for contacts and it is carried all the time – it follows its owner everywhere. Mobile communication is by nature very personal and person-to-person oriented. This, we believe, will make the mobile phone the de-facto device for the user to publish and manage presence information.

Presence enables a new delicate background communication layer and continuous mobile bonding based on up to date information. It also enables emotional bonding for virtual groups and acts as a tool for mobile virtual identity. People will have the ability to create content that they know is really useful to others – it is not just another Internet homepage that lacks visitors. The pages are actually “my and my friends' up-to-date personal pages, which I carry with me all the time”.

When presence information is added to contact information in the terminal, i.e. phonebook, it will create a dynamic entity that naturally extends and enriches the static representation of contacts. Presence enables more

private information sharing among small trusted groups as well as sharing your availability with a wider audience to enable more appropriate and effective communication. Presence information serves both rational as well as emotional and expressive needs.

Mobile presence is an efficient application in creating mobile awareness of close contacts. Users are always online and therefore it is not worth announcing this fact as is done with PCs. Context for mobile presence is real-life, not something artificial that you must log into. This is the fundamental difference to fixed Internet instant messaging.

In the mobile context, presence information is also likely to include information from the network, such as location, which has little meaning without mobility. Further, terminals in the mobile world differ from each other in terms of displays and other visual capabilities. When networks and terminals enable true mobile multimedia, it is also important to share information about terminal capabilities as one of the presence attributes. As there are legacy terminals that do not support the latest multimedia capabilities, this information is also important. In the PC environment, this is not an issue.

From availability to rich communication

Presence supports both rational and emotional user needs. There will always be users who only use the basic functionality that the presence service can offer – in these cases presence is used merely for availability sharing purposes. The availability update may be automated by the terminal or the network and only a minimal set of information is included in the presence view.

At the other end of the spectrum are people who view presence information as a means of sharing their own personality with their friends, family, colleagues and even strangers. In these cases users are willing to use personalised icons, messages and additional personal profiles.

Presence will be a pervasive part of daily mobile communication. Today, almost all Nokia mobile terminal users make voice calls, use the phonebook and most send text messages as well. Most of the users also use terminal profiles and personalise them. These are the mobile behaviours that will be among the first to benefit from presence information. A successful adaptation involves taking an application already widely used and enhancing it with presence information, ensuring fast and intuitive take-up of the service.

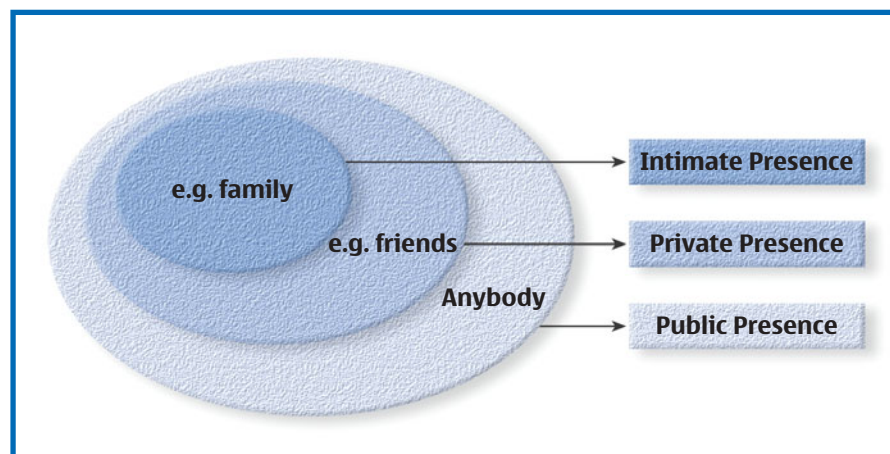
There are some rational reasons why presence will succeed in the mobile context. First, presence supports today's mobile communication and usage models, as described in the previous paragraph. Second, the usability will be as natural and automated as

possible. Nokia has a long history of researching easy usability and meeting the challenges. Third, presence can be all things – rational, fun and emotional – appealing to a wide range of users.

Maintaining privacy

A clear user need is the ability to share different status with different users. This leads to multiple layers of presence depending on the type of people who are viewing the information, as described in figure 1. While there is likely to be some very simple presence information that can be shared with anybody, users are not willing to share everything with everybody. There is also a need to share different information with different people. For example, users' availability might be different for co-workers and family. Further, location information is a typical example of information that people are likely to share only with a limited group of contacts that they have selected themselves. Whatever the information for whichever group, the user always has control over who is able to view and what they can see.

Figure 1. Multiple layers of presence.



Mobile presence usage scenarios

Mike and his problematic valves

Mike has recently started working for Pipemaster & Co. and today he is in the main district-heating centre for the first time. He is working on a valve control unit that he is unfamiliar with and thinks that more experienced people must have worked on this equipment. Mike decides to call for help. He looks at his colleagues' presence information from his terminal to see who is available for a quick consultation.

He has subscribed to all his co-workers' presence information in his presence enhanced phonebook. Once somebody changes his or her presence information, it is automatically updated in the background of Mike's terminal as well. This way he always has the latest presence information available. Susan is sick today, Peter seems to be travelling, but Bill seems to be at the office and available. Mike decides to make a quick call to him.

After the job Mike marks the task completed into a shared to-do list that is part of his presence information. He then moves on to the next job. There are problems again at the Ferry

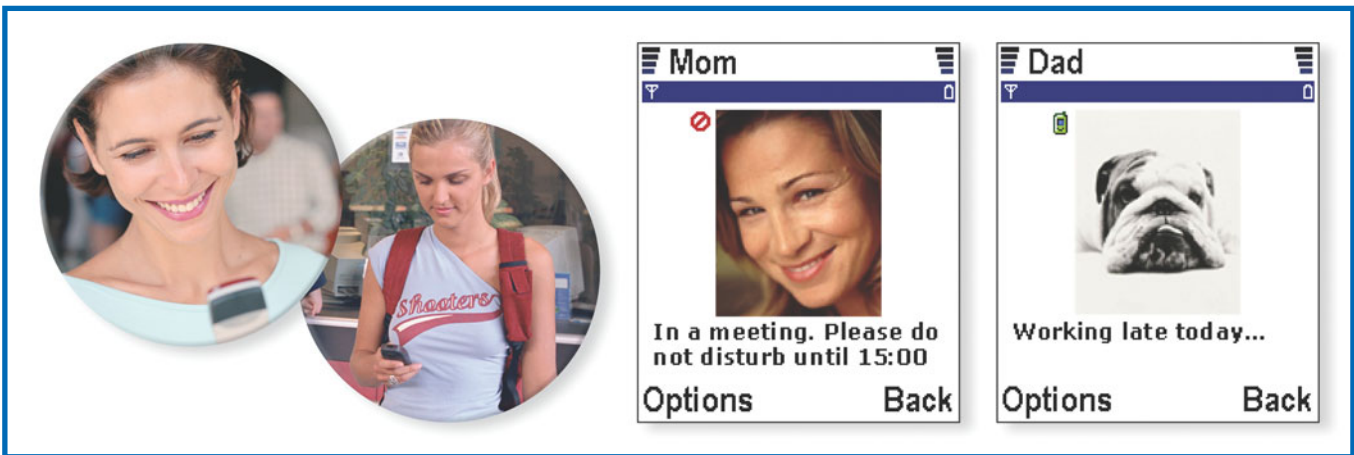
Island power plant. He indicates that he has accepted the job and heads there right away. At the same time he changes his preferred communication method to voice calls, as it is hard to read text-messages while driving. With a hands-free set Mike can still handle calls and people now know not to send him messages while he's driving to Ferry Island.

Lisa connecting with family

Lisa's mother is a busy professional woman with many meetings every day. Still, she and Lisa want to keep in touch. Lisa's school day is over and she switches on her mobile to be reachable. Her availability is seamlessly updated to her parents and friends' presence enhanced phonebook.

Lisa checks up on her mother's presence information from her mobile phone. The presence information is always up-to date and indicates that she is in a meeting and does not want to be disturbed. Lisa decides not to call her and sends a text-message instead. She knows that her mother will call her later when she's out of the meeting.

Looking at her dad's status, she sees something funny in his presence information today, an amusing picture and a slogan to match. She guesses that dad is not so happy about the situation and decides to call him.



Enhancing services with Presence

The basic end-user presence service establishes the basic publish-receive features of presence. This could be the core competence of mobile operators, as they have the ability to fetch various infrastructure related attributes, such as location, from their communication networks.

New presence-enabled services use presence information in dedicated application areas, such as real-time stock watch. This is a big opportunity for innovative entrepreneurial companies. Other examples may be network gaming services.

In addition, presence creates an alternative advertising and information-sharing channel. Subscribing to company presence information will give the users the ability to receive promotions and useful information directly into their phone. It won't be considered by the subscribers as spamming, since they have asked to receive that information on their terminals. Also, the advertisements won't disturb them since information is updated in the background, in contrast to SMS and MMS push. One example would be to enter a preferred airline into the phonebook. This company would be updating their presence information constantly with its latest offers and discounts. When the user sees an interesting flight, it would be easy to call the airline and book the trip.

With presence, team working is more efficient, bringing the ability to share information amongst the team members. More than their availability, they can share key data such as action plan status and sales figures.

Presence as an appealing business

The mobile domain, now reaching almost a billion subscribers worldwide, is a lucrative platform for new consumer services.

Presence can contribute to existing businesses and create a business of its own. There will be basic presence user-services, new presence enabled services as well as content and infrastructure variants of the presence business.

Operators and other service providers have a central role in mass adoption of presence services. The basic mobile presence service can have a central role in operators' service portfolio, since other services, i.e. the new presence services, can use the basic service.

A presence service encourages customer loyalty, as shown by instant messaging in the fixed Internet. Implementation of the basic presence service can also be a competitive advantage over other operators who are not implementing it: by binding their presence information to a particular operator's services, customers have high value services that other operators may not be able to offer without that information. Presence information sets up a very strong personal channel to individuals and an operator may also decide to sell presence information to other service providers with the user's consent.

As mentioned earlier, presence generates new traffic in the form of messaging, instead of unanswered calls. It also generates new voice and messaging traffic in the same way as fixed Internet instant messaging has already proved – a change in the

presence information may result in a greeting or other kind of message.

Operators can maximize the penetration of presence-enabled services among mobile users. The key things here are: Implementing open presence technology standards, promoting sales of the service, signing agreements with other operators on exchanging presence information and promoting innovative new presence enabled services. The ability to see presence information from other operator's networks is very important, as most users communicate with people beyond their home network.

The pricing of the presence service also needs to be considered carefully. Operators must ensure that their charging schemes encourage their users to take up the services and use them in a rich and meaningful way. We believe that there is considerable market potential in presence, comparable to SMS, if prices are set at a moderate level. Completely free presence services would not encourage a healthy presence industry, nor would they allow quality of service or new innovative presence services to be provided to end-users in the long run.

The importance of interoperability

The more presence capable terminals there are and the better the communication network coverage is for presence services, the more useful presence will be for end-users.

We believe presence will be pervasive in the future. Therefore we must ensure interoperability between different terminal manufacturers, different presence servers, and different mobile operators as well as between the fixed and mobile domains.

Instant Messaging and presence are moving from the desktop to the mobile domain. To support this, the Wireless Village initiative was set up by Ericsson, Motorola and Nokia, to build a community around new and innovative mobile Instant Messaging and Presence Services (IMPS). Through its supporters, the Wireless Village initiative aims to build a vibrant community of users and global business partners where the Internet and wireless domains converge.

Wireless Village will define a set of universal specifications for mobile IMPS, as well as procedures and tools for testing conformance and interoperability. The specifications will be used for exchanging messages and presence information between mobile devices, mobile services and Internet-based instant messaging services.

In short, Wireless Village is about building the market, making the specifications and ensuring interoperability. For more information, see www.wireless-village.org.

As technology standards and delivery platforms mature, new standards arise and become de facto. It is important that the infrastructure offering complies with these new standards. 3G networks and all-IP based mobile presence services are a clear continuance from the current presence infrastructure offering.

Summary

Availability of presence information will change mobile communication fundamentally. Users will be in touch with each other prior to initiating communication. Communication will become more effective and appropriate as users are able to share with others how and when they wish to be contacted.

Presence will also offer opportunities for personalisation. As the mobile device becomes a living extension and representation of one's identity and social network, it will play a more significant role in people's lives. Applications and services that enable people to express their emotions and aspirations have a big potential for success, given rapid implementation of presence infrastructure, open technologies, information sharing by operators, 3rd party software developers and moderate end-user pricing. With presence, every mobile terminal user becomes a content creator as well as a publisher.

Glossary

IMPS	Instant Messaging and Presence Services
SMS	Short Messaging Service
MMS	Multimedia Messaging Service

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